

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1520

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I **Thamaraiselvi S (192511042)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Thamaraiselvi S

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Analytical Problem Solving

1. Compare Type-1 and Type-2 Hypervisors for a Cloud Datacenter Hosting Mission-Critical Applications

Introduction

A hypervisor is virtualization software that allows multiple virtual machines (VMs) to run on a single physical server. There are two main types of hypervisors: **Type-1 (Bare-Metal)** and **Type-2 (Hosted)**. Selecting the appropriate hypervisor is important for achieving high performance, strong security, and efficient management in cloud datacenters that host mission-critical applications.

Type-1 Hypervisor

Type-1 hypervisors are installed directly on the physical hardware without requiring a host operating system. Since they interact directly with the hardware, they provide faster performance, lower latency, and better resource utilization. They are commonly used in enterprise cloud environments because they support advanced features such as live migration, high availability, disaster recovery, and centralized management. Examples include VMware ESXi, Microsoft Hyper-V, and Xen.

Type-2 Hypervisor

Type-2 hypervisors are installed on top of an existing operating system. They depend on the host OS for hardware access, which introduces additional overhead and reduces performance. Although Type-2 hypervisors are easy to install and useful for software testing, development, and learning, they are generally not suitable for production cloud datacenters because they provide lower performance and weaker security.

Performance Analysis

Type-1 hypervisors provide direct access to hardware resources, resulting in higher CPU efficiency, faster memory access, and improved storage performance. In contrast, Type-2 hypervisors must communicate through the host operating system, increasing resource consumption and reducing overall performance. Therefore, Type-1 hypervisors are better suited for applications requiring high availability and low response time.

Security Analysis

Type-1 hypervisors have a smaller attack surface because they do not rely on a host operating system. This minimizes the chances of malware or vulnerabilities affecting the virtual machines. Type-2 hypervisors inherit security risks from the host operating system, making them more vulnerable to attacks if the host OS is compromised.

Manageability Analysis

Type-1 hypervisors offer centralized management tools, automated VM provisioning, live migration, load balancing, and fault tolerance. These features simplify administration in large cloud environments. Type-2 hypervisors lack many enterprise management capabilities and are mainly intended for individual users or developers.

Conclusion

For mission-critical cloud datacenters, **Type-1 hypervisors are the preferred choice** because they provide superior performance, stronger security, higher reliability, better scalability, and easier management. They ensure continuous service availability and efficient utilization of cloud infrastructure.

2. Analyze How Virtualization Architecture Supports Workload Isolation and Identify Two Major Attack Surfaces

Introduction

Cloud service providers often host multiple organizations on the same physical server. For example, finance, healthcare, and student portal applications may run together while maintaining complete isolation. Virtualization architecture enables secure resource sharing by creating multiple independent virtual machines.

Virtualization-Based Isolation

Each virtual machine has its own operating system, virtual CPU, memory, storage, and network interface. The hypervisor ensures that the resources assigned to one VM cannot be directly accessed by another VM. This isolation prevents failures or attacks in one VM from affecting others and allows organizations to share physical hardware securely.

Benefits of Workload Isolation

Virtualization provides strong security, efficient resource utilization, simplified management, fault isolation, and support for regulatory compliance. Finance and healthcare

applications can securely process sensitive information while student portal services continue operating independently on the same server.

Attack Surface 1 – Hypervisor Attack

The hypervisor is the core component that manages all virtual machines. If attackers exploit a vulnerability in the hypervisor, they may gain control over every VM running on the server. This could lead to unauthorized access, data theft, and complete system compromise. Regular software updates, secure configuration, and hardware-assisted virtualization help reduce this risk.

Attack Surface 2 – VM Escape Attack

VM escape occurs when malicious software running inside a virtual machine breaks out of its isolated environment and accesses the host system or neighboring VMs. This attack threatens data confidentiality and system integrity. Strong VM isolation, timely security patches, and continuous monitoring help prevent VM escape attacks.

Conclusion

Virtualization architecture enables multiple organizations to safely share the same physical infrastructure while maintaining strong workload isolation. However, protecting the hypervisor and preventing VM escape attacks are essential for maintaining cloud security.

3. Analyze the VM Host Utilization Snapshot and Recommend Corrective Actions

Given System Status

- CPU Utilization = **92%**
- Memory Utilization = **88%**
- Storage I/O Latency = **35 ms**
- Average VM Boot Time = **170 seconds**

CPU Analysis

A CPU utilization of 92% indicates that the processor is heavily overloaded. The server has very little remaining processing capacity, causing virtual machines to compete for CPU resources. This results in slower application execution and reduced system responsiveness.

Corrective Actions:

- Add additional CPU cores.
- Distribute workloads across multiple hosts.
- Migrate heavily loaded VMs to less busy servers.
- Optimize applications consuming excessive CPU resources.

Memory Analysis

Memory utilization of 88% suggests that the server is approaching memory exhaustion. The operating system may begin swapping memory to disk, significantly reducing application performance.

Corrective Actions:

- Increase physical RAM.
- Enable memory optimization techniques.
- Remove unused virtual machines.
- Balance memory usage across multiple hosts.

Storage Analysis

Storage I/O latency of 35 milliseconds is much higher than the recommended value of below 10 milliseconds. High storage latency slows application performance and increases VM startup time.

Corrective Actions:

- Upgrade to SSD or NVMe storage.
- Improve storage caching.
- Optimize disk configuration.
- Separate high-I/O workloads.

VM Boot Time Analysis

An average boot time of 170 seconds is unusually high. This delay is caused by excessive CPU utilization, memory pressure, and slow storage access.

Corrective Actions:

- Upgrade hardware resources.

- Reduce storage bottlenecks.
- Optimize VM templates and startup services.

Conclusion

The primary bottlenecks are CPU overload, high memory utilization, and slow storage performance. Increasing hardware capacity, balancing workloads, and upgrading storage systems will significantly improve VM performance and reduce startup times.

4. Analyze How Software Defined Datacenter (SDDC) Improves Agility Compared with a Traditional Datacenter

Introduction

A traditional datacenter depends on manually configured physical servers, storage devices, and networking equipment. In contrast, a Software Defined Datacenter (SDDC) virtualizes compute, storage, and networking resources, enabling centralized and automated management.

Compute Virtualization

In a traditional datacenter, applications require dedicated physical servers, making deployment slow and resource utilization inefficient. SDDC uses virtual machines that can be created, migrated, and managed automatically, allowing rapid deployment and better hardware utilization.

Storage Virtualization

Traditional storage systems are managed individually, making expansion and backup complex. Storage virtualization combines multiple storage devices into a centralized storage pool. Features such as thin provisioning, snapshots, and automated backup improve flexibility and reduce administrative effort.

Network Virtualization

Traditional networks require manual configuration of routers, switches, and VLANs. SDDC introduces Software-Defined Networking (SDN), where administrators configure networks through software. This enables faster deployment, easier security policy implementation, and simplified network management.

Advantages of SDDC

- Faster application deployment.
- Automated infrastructure management.
- Better resource utilization.
- Improved scalability.
- Reduced operational costs.
- High availability and disaster recovery support.

Conclusion

Software Defined Datacenters provide greater agility than traditional datacenters by virtualizing compute, storage, and networking resources. Automation and centralized management allow organizations to respond quickly to changing business requirements while improving efficiency and reducing costs.

5. Analyze Identity Mismatches in Multi-Cloud Federation and Propose a Secure Identity Design

Introduction

Organizations increasingly use multiple cloud providers such as AWS, Microsoft Azure, and Google Cloud. Since each provider has its own identity management system, users often experience authentication failures due to identity mismatches. A secure identity federation model is required to enable seamless access across multiple cloud platforms.

Problem Analysis

Identity mismatches occur because cloud providers use different authentication methods, user directories, and access policies. As a result, users may require multiple accounts and passwords, increasing administrative complexity and security risks.

Identity Federation Solution

Identity federation establishes trust between cloud providers using a centralized Identity Provider (IdP). Users authenticate once with the Identity Provider, which securely verifies their identity and grants access to authorized cloud services.

Single Sign-On (SSO)

Single Sign-On allows users to log in once and access multiple cloud platforms without repeated authentication. This improves user convenience while reducing password-related security issues.

Federation Standards

Modern federation solutions use standard authentication protocols such as:

- Security Assertion Markup Language (SAML)
- OAuth 2.0
- OpenID Connect (OIDC)

These protocols enable secure identity exchange between different cloud providers.

Privacy and Security Measures

To protect user identities:

- Encrypt identity information during transmission.
- Apply Multi-Factor Authentication (MFA).
- Implement Role-Based Access Control (RBAC).
- Share only necessary user attributes.
- Maintain audit logs for monitoring and compliance.

Conclusion

A secure multi-cloud federation architecture based on Identity Providers, Single Sign-On, SAML, OAuth 2.0, and OpenID Connect ensures secure authentication, protects user privacy, simplifies identity management, and enables seamless access across multiple cloud platforms.